

Fall 2026
CS1660/2060 Introduction to Cloud Computing
Wednesday 6:00 – 8:20 PM EST

COURSE DESCRIPTION

This course provides students with a comprehensive understanding of cloud computing technologies and their practical applications. Topics include virtualization, containerization (Docker, Docker Compose), container orchestration (ECS/Fargate, Kubernetes), serverless computing (Lambda, API Gateway), authentication (OAuth2, Cognito), cloud storage (S3, DynamoDB), and infrastructure as code (Terraform). The course culminates in a capstone project: students build and deploy an LLM-powered SaaS application on AWS.

The course covers the design principles and architectures of cloud computing, including infrastructure, platform, and software as a service models. Students will explore public cloud services, data center architecture, virtualization, and distributed processing frameworks.

The course is a mix of lectures and hands-on labs. Students are responsible for readings, completing labs, and assignments. The final grade is based on assignments (40%), in-class quizzes (20%), and a final project (40%). Course prerequisite is CS 1550 – Introduction to Operating Systems, or equivalent.

COURSE TOPICS

- Introduction to Cloud Computing and AWS
- Virtualization (hypervisors, Multipass, cloud-init, EC2)
- HTTP, APIs, and Client-Server Architecture
- Containerization (namespaces, cgroups, Docker, Compose)
- Container Orchestration (ECS/Fargate, Kubernetes)
- Cloud Storage and Hosting (S3, CloudFront, DynamoDB)
- Event-Driven Architecture (SNS, SQS, EventBridge)
- Serverless Computing (Lambda, API Gateway, VPC Link)
- Identity and Access Management (IAM, OAuth2/OIDC, Cognito)
- Cloud Observability (CloudWatch, Prometheus, load testing)
- Infrastructure as Code (Terraform, threaded throughout labs)

COURSE PREREQUISITES

Successful completion of CS 1550 – Introduction to Operating Systems, or equivalent course.

REQUIRED READING

Operating Systems: Three Easy Pieces by Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau. Available free online at <https://pages.cs.wisc.edu/~remzi/OSTEP/>. Selected chapters on virtualization will be assigned as pre-class readings.

COURSE INSTRUCTOR

Dan Mahoney, Primary course instructor
Email: dpm79@pitt.edu

COURSE DELIVERY

In-person lectures with online platforms (Canvas, GitHub Classroom). We will be using **Amazon Web Services (AWS)** for assignments and projects. Lectures follow a two-block format: two approximately 55-minute sessions separated by a 15-minute break.

AWS FREE TIER INFORMATION

Students will use their own AWS accounts. As of 2026, AWS offers:

- **Free Plan:** New accounts receive \$100–\$200 in credits, valid for 6 months
- **Always Free:** Ongoing free tier for EC2 (750 hrs/mo t3.micro), Lambda (1M requests/mo), S3 (5GB), DynamoDB (25GB), and more

Important: Stop EC2 instances when not in use. Set up billing alerts at \$5. Use us-east-1 region for all assignments.

OFFICE HOURS AND COMMUNICATION

Office hours: Wednesday 2:00–5:00 PM EST in Sennott Square 6510, or by Zoom appointment.

Email questions to the instructor with subject line beginning "CS1660:" or use the class Discord server.

IMPORTANT DATES

Add/drop deadline: See academic calendar
Thanksgiving recess: November 22–30, 2026
Final project deadline: Finals week

COURSE EVALUATION & ASSESSMENT

Students are encouraged to attend class regularly, complete assigned readings, and participate in discussions.

ASSIGNMENTS	QUIZZES	FINAL PROJECT
40%	20%	40%

Assignments: Assignments released throughout the semester, submitted via GitHub Classroom. CI pipelines grade submissions automatically.

Quizzes: Four in-class, paper-based quizzes covering lecture material and assigned readings. Quizzes are closed-book and require attendance.

Final Project (LLM SaaS): All students complete the same capstone project: an LLM-powered SaaS application deployed on AWS. The stack includes a static chat frontend (S3/CloudFront), API Gateway with Cognito authentication, and a Fargate-hosted backend running a FastAPI proxy and vLLM inference sidecar, with DynamoDB for usage metering. Infrastructure is defined in Terraform. Local development uses Docker Compose for parity with the deployed environment. CS 1660 and CS 2060 students build the same project; grading rigor differs. Project releases in week 10 and is due during finals week. Deliverables include a working deployment, a load test, and a recorded video demonstration.

GRADE SCALE

A+ 97–100%	A 93–96%	A- 90–92%
B+ 87–89%	B 83–86%	B- 80–82%
C+ 77–79%	C 73–76%	C- 70–72%
D+ 67–69%	D 63–66%	D- 60–62%
F <60%		

LATE POLICY

Students receive 48 hours of no-questions-asked late time for the semester. After exhausting this reserve, submissions are penalized 10% per day. Extensions beyond this require documentation of medical or family emergency.

AI USAGE POLICY

AI tools (ChatGPT, Claude, GitHub Copilot, etc.) are permitted in this course with the following guidelines:

- **Permitted uses:** Debugging code, explaining error messages, clarifying concepts, generating boilerplate, and brainstorming approaches.

- **Your responsibility:** You must understand and be able to explain any code or content you submit. If you cannot explain your submission, it will be treated as a potential academic integrity violation.
- **Not permitted:** Using AI to complete entire assignments without understanding the solution, or using AI during in-class quizzes.
- **Acknowledgment:** If AI significantly contributed to your solution, note this in your submission (e.g., "Used Claude to debug Docker networking issue").

The goal of this course is to learn cloud computing, not to test your ability to avoid AI. Use these tools as you would in a professional setting—to augment your abilities, not replace your understanding.

COURSE SCHEDULE

Schedule is approximate and may be adjusted. Refer to Canvas for current dates.

Week	Block A (6:00–6:55)	Block B (7:10–8:20)
Week 1 8/26	Cloud concepts, why cloud is hard	AWS tour, billing alarms, GitHub Classroom
Week 2 9/2	Virtualization: hypervisors, VM anatomy	Multipass lab: launch, cloud-init, exec
Week 3 9/9	EC2 deep dive, security groups, pricing	HTTP, APIs, client-server, curl
Week 4 9/16	Container primitives: namespaces, cgroups	Docker: images, run, exec, lifecycle
Week 5 9/23	Docker Compose + ECR	ECS/Fargate I: task definitions, containers
Week 6 9/30	ECS/Fargate II: services, ALB, task roles	API Gateway, VPC Link, authorizers
Week 7 10/7	Lambda	IAM: policies, roles, task roles
Week 8 10/14	OAuth2, OIDC, JWT anatomy	Cognito: hosted UI, integration demo
Week 9 10/21	S3 + CloudFront static hosting	DynamoDB: tables, queries, single-table design
Week 10 10/28	Final project kickoff: full architecture	Terraform consolidation: state, modules
Week 11 11/4	Event-driven architecture: SNS, SQS	Kubernetes I: concepts, pods
Week 12 11/11	Kubernetes II: deployments, services	Kubernetes III: networking, ingress, scaling
Week 13 11/18	Observability: CloudWatch, metrics	Live load test of project deployments

— 11/25	<i>Thanksgiving Recess</i>	<i>No class</i>
Week 14 12/2	Project Demo Day	Project Demo Day (continued)
Finals 12/9	Final Exam (during normal class time, 6:00–8:20 PM)	

ASSIGNMENT POLICY

- Check Canvas regularly for announcements, schedules, and materials.
- Assignments must be submitted via GitHub (main branch) and Canvas before the deadline.
- Assignments are typically due two weeks after release unless otherwise noted.
- Students are encouraged to discuss concepts but must complete work independently.

ACADEMIC INTEGRITY

Students must comply with the University of Pittsburgh's Policy on Academic Integrity. Unless explicitly stated, all work must be completed independently. Discussion of concepts is encouraged; sharing solutions is not permitted. Violations will result in a minimum penalty of zero on the assignment and may be reported to the Dean's office.

DIVERSITY AND INCLUSION

Everyone is welcome, and all members of the classroom are expected to act respectfully.

RELIGIOUS OBSERVANCES

To accommodate religious holidays, inform the instructor by email within the first two weeks of the term of any conflicts with scheduled class activities.